Context Before Synthesis: A Pre-Registered Study of Generative Augmentation for Rare-Arrhythmia Detection in Unseen Patients

Course project report, 10-page version

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Abstract

Beat classifiers trained on the MIT-BIH Arrhythmia Database detect normal and ventricular beats well, but in patients absent from training they miss most supraventricular (S) and fusion (F) beats, the rare classes with clinical weight. Class-conditional generative models promise to supply the missing rare beats. We tested that promise in a pre-registered study under the inter-patient protocol: selection rules fixed before the experiments, every choice made on validation patients, and the 22 test patients read once. Across two classifiers and a 96-run grid, neither a conditional VAE nor a WGAN-GP detectably improved the rare-class macro-F1 over no augmentation (best difference +0.025, 95% CI −0.018 to +0.047) or over class weighting, random oversampling and SMOTE, and no synthetic-quality metric predicted the gain: the synthetic beats reproduced the few training patients they were learned from. Twenty-three follow-up analyses, each registered before it ran, show what does generalise, namely patient context. Expressing each beat relative to the patient's own beats, judging prematurity against the patient's mean rhythm, vetoing S calls during atrial fibrillation, training with patients from two further databases and averaging ten runs raised the test score from 0.113 to 0.587 (S F1 0.647, F F1 0.527), above two classic inter-patient methods on S and V beats and above a recent one on F beats. On 977 patients no analysis had used, the pipeline's S F1 was 0.294 to 0.414. Evaluated on already-seen patients, the same network would have scored 0.837 instead of 0.088.

1 Introduction

Ambulatory monitors record the ECG for one or more days [21]; at 70 beats per minute a day holds about 100,000 beats, far more than anyone reads. Software therefore classifies every beat and flags the rare ones: supraventricular ectopic (S) beats, whose burden predicts atrial fibrillation and stroke [3]; ventricular ectopic (V) beats, whose burden goes with a weakened left ventricle [2]; and fusion (F) beats. The standard benchmark is the MIT-BIH Arrhythmia Database [14] with the beat classes of the AAMI standard [1].

The hard part is the patient. When the test patients are kept out of training, as de Chazal et al.'s inter-patient protocol [5] requires and as a clinic faces, published methods detect S beats with an F1 of 0.511 to 0.686 [5, 10, 24], and F beats, where reported, with 0.477 [24]. Many studies instead split beats at random, so that every patient appears in both training and test, and report far higher scores [11]; those scores largely measure how well a model recognises patients it has already seen.

Rare beats are scarce and concentrated. The training split holds 644 S and 388 F beats against 35,496 normal ones, and in the test split one patient carries 1,381 of the 1,836 S beats and another 362 of the 388 F beats. Class-conditional generative models, conditional VAEs [20] and Wasserstein GANs [8], can synthesise convincing rare beats, and augmentation with GANs and diffusion models has been reported to improve MIT-BIH classifiers [18, 11], mostly on random splits. We asked whether the remedy survives the patient boundary, and registered the question before reading any test data:

> *With training and test patients strictly apart, does class-conditional generative augmentation improve detection of the rare classes S and F, compared with no augmentation, a class-weighted loss, random oversampling and SMOTE?*

Every rule that chose a model, a variant or a strength was fixed in advance and applied to validation patients at a gate closing each phase; the test file opened once, for the final evaluation. In all, 761 runs are logged. The paper makes four contributions:

1. **A pre-registered negative result.** Generative augmentation did not detectably improve the detection of rare arrhythmias in unseen patients over three conventional remedies for class imbalance, and synthetic-quality metrics did not predict its usefulness; a nearest-patient analysis explains why (Sections 5.1 and 5.2).

2. **A patient-context pipeline** that raises the rare-class macro-F1 on the inter-patient benchmark from 0.113 to 0.587, above two classic methods on S and V beats and above a recent one on F beats and on the mean of S and F (Section 5.3), with its generalisation tested on 977 unseen patients (Section 5.4).
3. **A same-pipeline measurement of patient overlap**: the rare-class macro-F1 rises from 0.088 to 0.837 when the test patients were seen in training, consistently over three random splits (Section 5.5).
4. **An explainable system**: per-segment saliency and notes written by a language model from the classifier's output alone, under a fact checker that rejects any claim not in that output (Section 5.6).

2 Related work

Evaluation. The AAMI EC57 standard [1] groups the MIT-BIH labels into the classes N, S, V, F and Q and asks for record-by-record reporting; de Chazal et al. [5] split the records into DS1 for training and DS2 for testing so that no patient is in both. Mousavi et al. [15] report the same model under both protocols and find the random split far easier; we measure the gap on one pipeline.

Rhythm and patient context. The strongest inter-patient methods rely on RR intervals measured against the patient's own rhythm: Lin and Yang [12] divide them by the recording's mean, Huang et al. [10] call a beat S when it comes early against that mean, and Zhu et al. [24] feed RR intervals to a convolutional network. Our patient-relative beat extends the same idea from timing to morphology.

Generative augmentation of ECG. GANs [18] and conditional diffusion models [11] have been used to augment MIT-BIH classifiers, and SimGANs [6] constrain a GAN with McSharry et al.'s ECG simulator [13]. We compare conditional VAEs [20] and WGAN-GPs [8] with SMOTE [4], random oversampling and a class-weighted loss under the inter-patient protocol, and test diffusion and the simulator loss as follow-ups.

Explanations. We compare Grad-CAM [17], integrated gradients [22], SmoothGrad [19] and occlusion by faithfulness, and restrict language-model notes to the classifier's output by a fact checker.

3 Problem setup and protocol

Data. The MIT-BIH Arrhythmia Database [14, 7] holds 48 half-hour two-lead recordings whose beats were labelled by cardiologists. We use lead MLII at 360 Hz, map the labels to the AAMI classes N, S, V and F, and exclude the four paced records and the few Q beats. The 22 DS1 records give training (17) and validation (5) patients and the 22 DS2 records the test patients (Table 1); generators, oversamplers and normalisation see training patients only. Figure 1 shows the classes: an S beat comes early but looks almost like the patient's normal beat, a V beat is wide and different, an F beat lies between the two.

Split	Records	N	S	V	F
training	17	35,496	644	3,030	388
validation	5	10,370	300	758	27
test (DS2)	22	44,259	1,837	3,221	388

Table 1: Records and beats per split. S beats are under 4% of each split, and F beats at most 1%.

Measure. For class c with sensitivity Se_c and positive predictivity PPV_c, $\mathrm{F1}_c = 2\,\mathrm{Se}_c\mathrm{PPV}_c/(\mathrm{Se}_c + \mathrm{PPV}_c)$. The primary measure is the rare-class macro-F1 on the test patients,

$$M = \tfrac{1}{2}\left(\mathrm{F1}_S + \mathrm{F1}_F\right), \tag{1}$$

with 95% confidence intervals from 10,000 bootstrap resamples of test records (patients), paired differences between conditions, and Holm's correction [9] within each family of comparisons; the confirmatory family holds the four comparisons of a generative condition with no augmentation.

Pre-registration and test lock. Every selection rule and hypothesis was written down before the experiment it governs, and each phase ended with a gate at which the rule was applied and approved. The test array can only be read through a guarded loader that the final evaluation alone unlocks, and every read is logged. The follow-up analyses were registered as new versions of the rules (21 in all) before they ran, are reported apart from the primary results, and are labelled post hoc.



Figure 1: The beat classes on the ECG (training records only). The RR intervals (arrows) reveal an early beat; the rare beats of each patient differ from the same patient's normal beat in their own way.

4 Methods

4.1 The pre-registered pipeline

Each stage was chosen on validation patients by a registered rule (Figure 2, left). **Beat.** A 0.5 s window around the reference R peak (180 samples), z-scored per beat, with three RR features as side input: the intervals before and after the beat and the ratio of the former to the mean of the previous ten (the best of four representations). **Classifiers.** A shallow CNN (73k parameters) and a BiLSTM (135k), the best of four architectures, trained with three seeds and early stopping on validation S F1 (validation holds too few F beats to guide selection); pretraining on PTB-XL did not help. **Generators.** A conditional VAE and a WGAN-GP, each the best of four designs in a pilot, trained on training patients only with three seeds into frozen banks of synthetic beats. **Augmentation.** Each rare class c with n_c real training beats is brought up to a fraction $s \in \{0.1, 0.25, 0.5\}$ of the normal count n_N with $m_c = s\,n_N - n_c$ synthetic beats, taken from $2m_c$ bank beats as the m_c closest to real training beats by dynamic time warping [16]; s is chosen per condition on validation. With no augmentation, a class-weighted loss, random oversampling and SMOTE as competitors, this is a grid of 96 runs (2 classifiers, 3 seeds, no augmentation and 5 conditions at 3 strengths). **Training.** Cross-entropy (class-weighted only in that condition), AdamW with weight decay 0.01, batches of 256, a learning rate of 10^{-3} or $3 \cdot 10^{-4}$ chosen on validation, at most 40 epochs, and early stopping with patience 5 on validation S F1, never before epoch 10; every condition has the same budget.

4.2 The patient-context pipeline

The follow-up analyses replaced population-level inputs with patient-relative ones (Figure 2, middle). For beat b with z-scored window x_b and R peak at time τ_b, the patient template is the element-wise median of the windows of every beat of the same recording within 150 s, and the classifier sees the deviation from it:

$$\tilde{x}_b = x_b - t_b, \qquad t_b = \underset{j:\,|\tau_j - \tau_b| \le 150\,\mathrm{s}}{\mathrm{median}} x_j. \tag{2}$$

Figure 3 shows the effect: the shape of a normal beat differs from patient to patient (record 118 has a right bundle branch block), but after the template is subtracted a normal beat is close to zero everywhere and only the deviation of an ectopic beat from that patient's usual beat remains.

RR intervals are divided by the mean RR interval $\overline{RR}$ of the recording, $\tilde{r}_b = RR_b^{\mathrm{prev}}/\overline{RR}$. Neither step uses a label. Ten independently trained runs are averaged, $\bar{p}(c \mid b) = \frac{1}{10}\sum_{k=1}^{10} p_k(c \mid b)$, and a beat the ensemble

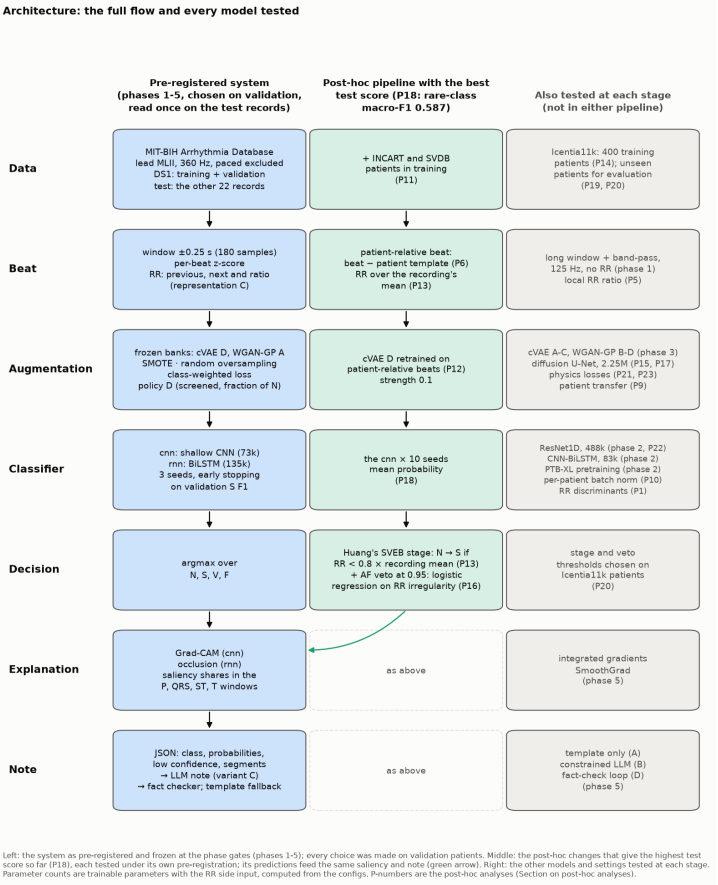


Figure 2: The pipeline. Left: the pre-registered system. Middle: the patient-context pipeline with the best test score (Section 5.3). Right: what else was tested at each stage.

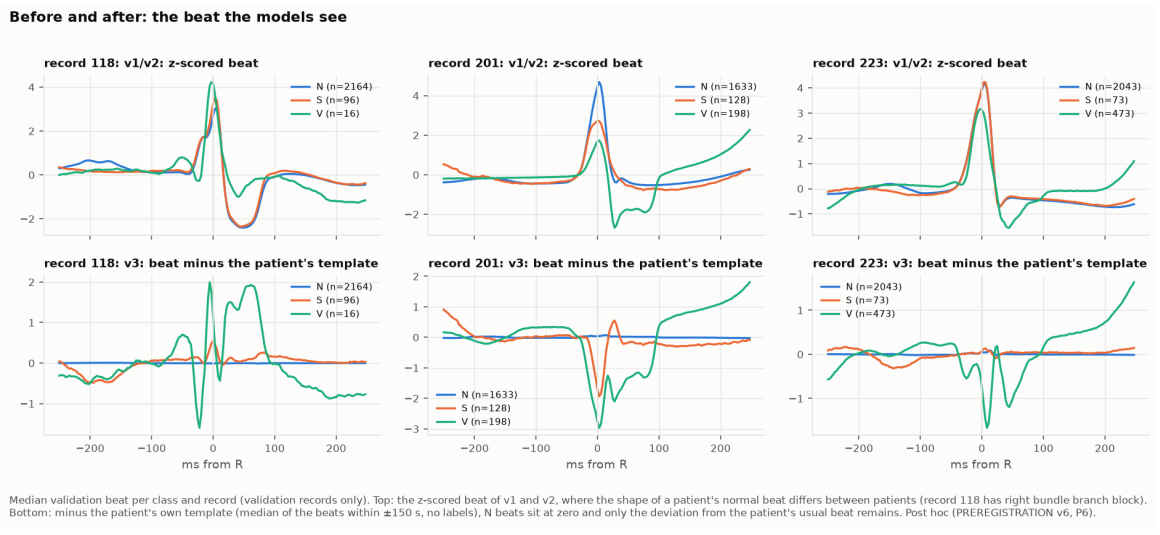
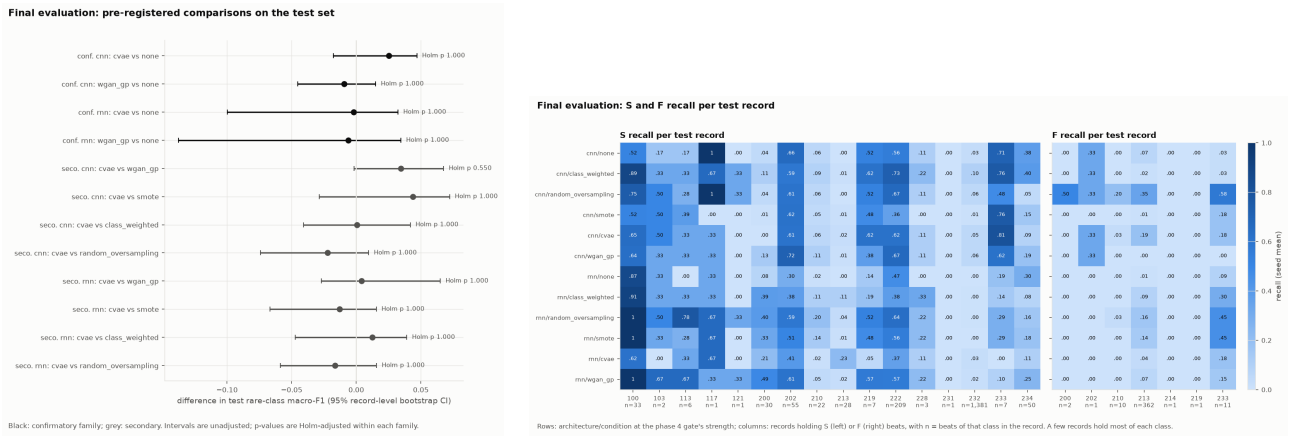


Figure 3: What the classifier sees, for three validation patients (median beat per class). Top: z-scored beats, whose shape differs between patients. Bottom: the same beats minus each patient's template (Equation 2); a normal beat is close to zero and an ectopic beat keeps only its deviation from the patient's usual beat.



(a) (b)

Figure 4: The pre-registered comparison on the test patients. (a) Differences in rare-class macro-F1 with 95% intervals and Holm-adjusted p: black, the confirmatory family; grey, the secondary one. (b) S and F recall per test record for all twelve models: two records (232 for S, 213 for F) hold most rare beats, and every model misses most of them.

calls normal becomes S when it is premature, unless an atrial-fibrillation detector fires:

$$\hat{y}_b = S \quad \text{if} \quad \arg\max_c \bar{p}(c \mid b) = N, \;\; \tilde{r}_b < 0.8 \;\; \text{and} \;\; a_b < 0.95, \tag{3}$$

and $\hat{y}_b = \arg\max_c \bar{p}(c \mid b)$ otherwise. The threshold 0.8 is Huang et al.'s [10], not tuned; a_b is the atrial-fibrillation probability of a logistic regression on five RR-irregularity measures, fitted on the 25 records of the MIT-BIH Atrial Fibrillation Database [7] (other subjects), with its threshold chosen on training-side data. Training also adds the patients of two further databases (St Petersburg INCART and the MIT-BIH Supraventricular Arrhythmia Database [7]), and the generators are retrained on patient-relative beats.

4.3 Explanations and notes

Saliency is summarised as the share of attention in fixed P, QRS, ST and T windows around the R peak. A language model (Gemini) writes a short note from a JSON holding only the prediction, the probabilities, a low-confidence flag and the two most salient segments; a fact checker rejects any number, class, segment or finding not in that JSON and falls back to a template note.

5 Results

5.1 The pre-registered answer: synthetic beats did not help

Table 2 gives the single test read and Table 3 the registered hypotheses. For the CNN, no augmentation scored 0.088, the cVAE 0.113 and the WGAN-GP 0.079, against 0.113 for the class-weighted loss and 0.136 for random oversampling. The cVAE's gain, +0.025 (95% CI −0.018 to +0.047), points the right way and nearly tripled the CNN's sensitivity for F beats (0.064 to 0.185), but no confirmatory comparison survived Holm's correction (all adjusted $p = 1.000$), with or without record 202, whose patient also appears in training as record 201. Against the conventional remedies the cVAE was level: +0.001 against the class-weighted loss and −0.022 against random oversampling. For the BiLSTM all differences were smaller still (Figure 4a). On two other databases, St Petersburg INCART and the MIT-BIH Supraventricular Arrhythmia Database, generative augmentation was detectably better than no augmentation in 1 and worse in 1 of 8 comparisons. Normal and ventricular beats were detected at about the published level (CNN with the cVAE: N recall 0.938, V F1 0.893); S and F beats were not.

Where the errors are. Two test patients set most of the primary measure (Figure 4b): across the twelve models, S recall on record 232 ranged from 0.00 to 0.10, and F recall on record 213 from 0.00 to 0.35. The CNN without augmentation labelled 1,367 of the 1,836 S beats and 333 of the 388 F beats as normal, and the S beats it did find were more premature than those it missed (median ratio of the previous RR interval to the local mean 0.57 against 0.72). The model found the S beats that broke the rhythm most and missed those close to it, which pointed the follow-up analyses to judging prematurity against each patient's own rhythm.

Architecture	Condition	Strength	Rare macro-F1 [95% CI]	N recall
cnn	none	–	0.088 [0.037, 0.194]	0.962
cnn	class weighted	0.5	0.113 [0.057, 0.207]	0.933
cnn	random oversampling	0.1	0.136 [0.061, 0.247]	0.844
cnn	smote	0.1	0.070 [0.025, 0.172]	0.985
cnn	cvae	0.25	0.113 [0.049, 0.217]	0.938
cnn	wgan gp	0.1	0.079 [0.033, 0.164]	0.955
rnn	none	–	0.075 [0.014, 0.232]	0.961
rnn	class weighted	0.75	0.061 [0.022, 0.168]	0.865
rnn	random oversampling	0.1	0.090 [0.028, 0.196]	0.883
rnn	smote	0.1	0.086 [0.025, 0.206]	0.894
rnn	cvae	0.1	0.073 [0.035, 0.157]	0.926
rnn	wgan gp	0.5	0.069 [0.027, 0.156]	0.857

Table 2: The pre-registered comparison: the 22 test patients, read once. Rare-class macro-F1 with 95% intervals from a bootstrap over patients, and the recall of normal beats; the strength s is the fraction of the N count chosen on validation.

Hypothesis	Statement	Verdict	Evidence
H1	confirmatory: a generative condition improves test rare-class macro-F1 over none (Holm, 4 tests)	not supported	no confirmatory comparison significant
H1b	exploratory: gains larger for F than for S	mixed	F gain larger than S gain in 2 of 4 generative conditions
H1c	secondary: generative beats SMOTE, random oversampling and class-weighted loss	not supported	0 of 6 comparisons of the best generative condition with a baseline significant
H2	exploratory: intra-patient evaluation inflates the baseline and the gains	not consistent	inflation of the baseline +0.748; larger intra-patient gain for 0 of 2 generators
H3	exploratory: a quality metric predicts the downstream gain	not supported	0 of 5 metrics with an interval excluding zero
H4	exploratory: augmented models are at least as faithful as the baseline	mixed	augmented faithfulness at least the baseline's in 1 of 4 model pairs

Table 3: The registered hypotheses, their verdicts (the rules applied as written) and the evidence.

5.2 Why the synthetic beats did not transfer

Before use, the quality metrics were checked on deliberately degraded beats: dynamic time warping and a spectral distance ranked them correctly, a Fréchet-distance analogue and train-on-synthetic-test-on-real did not. Five metrics covering fidelity, diversity, memorization and a classifier two-sample test were then compared with the downstream gain over 96 checkpoints of 24 generator runs; none predicted it (Spearman ρ from -0.20 to $+0.18$, every interval including zero). The beats themselves show why. Between 57% and 65% of the synthetic S beats lie nearest to a single training patient, record 209, just as 60% of the real training S beats do, while the validation patients' S beats lie nearest to training patients who contribute only 1–2% of them. The generators reproduce the training distribution faithfully, and that distribution is dominated by one patient; a new patient's rare beats are not in it (Figure 5).

5.3 Patient context lifts detection above published results

Figure 6 traces the follow-up analyses on the test patients. Patient-relative beats raised the CNN without augmentation from 0.088 to 0.371 (P7), the largest single step. Generators retrained on patient-relative beats lifted the cVAE condition to 0.368 (P12). Normalising RR intervals by the recording's mean with Huang et al.'s rule on top raised it to 0.523 (P13); the atrial-fibrillation veto raised every condition by 0.018 to 0.034, each 95% interval above zero (P16); and averaging ten runs gave 0.587 [0.086, 0.702], with S F1 0.647 and F F1 0.527 (P18). Within this pipeline the cVAE scored highest, 0.046 to 0.271 above the other conditions, mostly through fusion beats, but none of these differences is detectable with 22 test patients (Holm p from 0.107 to 1.000): synthesis on top of context is a promising point estimate, not a finding.

Table 4 sets the pipeline beside published results. It is above the two classic methods on both classes they report, S F1 0.647 against 0.511 and 0.577 and V F1 0.936 against 0.797 and 0.924, and above Zhu et al.'s recent method on fusion beats (0.527 against 0.477) and on the mean of S and F (0.587 against 0.582), while Zhu et al. remain ahead on S (0.686) and V (0.960) beats. Two qualifications apply: our pipeline was trained with additional patients from two further databases, and, built after the final evaluation, it read the same test patients more than once.

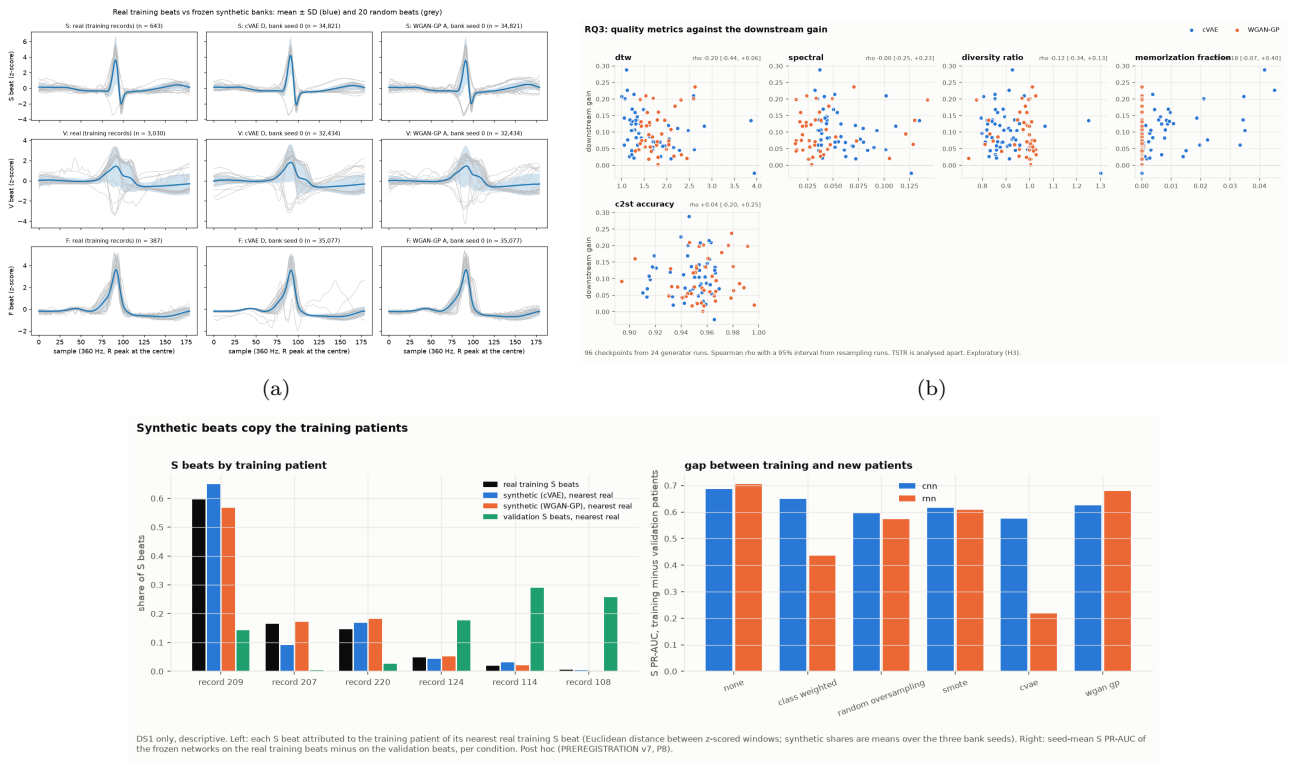


Figure 5: Why the synthetic beats did not transfer. (a) Real training beats beside the frozen cVAE and WGAN-GP banks (mean ± SD and 20 random beats): the synthetic beats look right. (b) No quality metric predicts the downstream gain (96 checkpoints of 24 generator runs). (c) Each S beat attributed to its nearest training patient: synthetic beats follow the real training mix, dominated by record 209, while the validation patients' S beats are nearest to patients the training set barely holds.

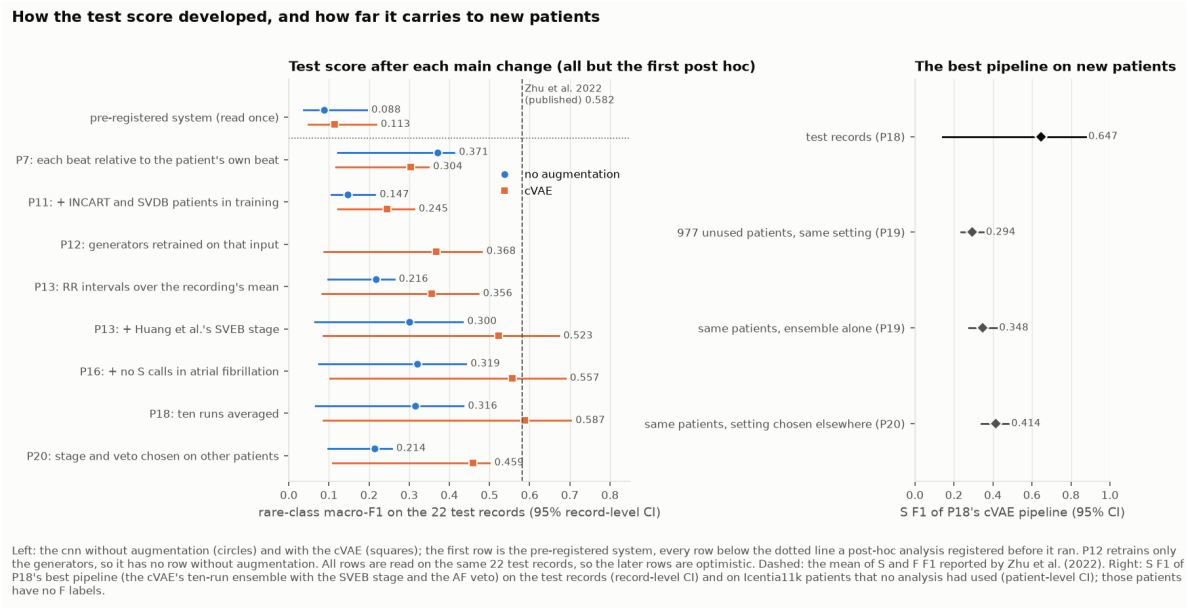


Figure 6: The test score after each main change, without augmentation (circles) and with the cVAE (squares); every row below the dotted line is a follow-up analysis; dashed, Zhu et al.'s published mean of S and F F1. Right: the best pipeline's S F1 on the test patients and on unseen Icentia11k patients.

	S F1	V F1	F F1	Mean of S and F
Test patients unseen in training (DS2)				
de Chazal et al. 2004 [5]	0.511	0.797	–	–
Huang et al. 2014 [10]	0.577	0.924	–	–
Zhu et al. 2022 [24]	0.686	0.960	0.477	0.582
Mousavi et al. 2019[†] [15]	0.907	0.997	–	–
Ours, pre-registered (CNN with the cVAE)	0.173	0.893	0.054	0.113
Ours, patient-context pipeline[‡]	0.647	0.936	0.527	0.587
Test patients also seen in training (beats split at random)				
Kwak and Jung 2024 [11]	0.947	0.985	0.903	0.925
Ours, the same CNN (no augmentation)				0.837

Table 4: Beside published results (F1 computed from the reported sensitivity and positive predictivity where a paper gives no F1). [†]Far above every other inter-patient report; its protocol could not be checked here. [‡]Post hoc, and trained with patients of two further databases as well as DS1.

Setting	Variant	Rare-class macro-F1 [95% CI]
Huang's threshold 0.8, AF veto 0.95 (P18)	no augmentation	0.316 [0.066, 0.434]
	WGAN-GP	0.474 [0.081, 0.604]
	diffusion model	0.433 [0.109, 0.554]
	cVAE	**0.587** [0.086, 0.702]
Thresholds chosen on other patients (P20)	cVAE	0.459 [0.110, 0.500]
	cVAE with a simulator (Euler) loss (P21)	0.481 [0.106, 0.529]
	cVAE with a fitted simulator loss (P23)	0.465 [0.112, 0.506]
	ResNet1D with the cVAE (P22)	0.486 [0.149, 0.529]

Table 5: Other follow-ups in the patient-context pipeline (ten-run ensembles, the 22 test patients, post hoc). Against the cVAE with the same thresholds, none of the lower block is detectably different (Holm p 0.24 to 0.62).

What else was tried. Table 5 lists the other ideas tested inside the same pipeline. A class-conditional diffusion model and the WGAN-GP stayed below the cVAE, and with the thresholds chosen on other patients neither a physics prior from McSharry et al.'s simulator [13], as in SimGANs [6], nor a ResNet1D with 6.7 times the CNN's parameters detectably improved on the cVAE.

5.4 Generalisation to unseen patients

To discount those repeated reads, the frozen pipeline was run on 977 patients of the Icentia11k database [23] that no analysis had used (S and V labels only; Icentia11k has no F label). Its S F1 was 0.294 [0.237, 0.350] with its own thresholds, and 0.414 when the two thresholds were chosen on 388 other Icentia11k patients; on the test records that choice gave 0.459, because the test record that holds most S beats resembles no tuning patient. Patient context therefore transfers, but less than the test score suggests, and the cVAE was level with the other conditions there (differences -0.001 to $+0.008$, none detectable).

5.5 Patient overlap inflates the score almost tenfold

Trained and tested on a random beat split of all 44 records, the same CNN without augmentation reached 0.837 ± 0.026 (Figure 7), and 0.803 and 0.777 on two further random splits, against 0.088 on unseen patients. Generative augmentation helped under neither protocol (with patients seen in training: cVAE -0.017, WGAN-GP -0.052). Random-split results above 0.9 belong beside our own 0.837, not beside inter-patient results (Table 4).

5.6 A prediction a clinician can check

Of the four saliency methods, the registered rule chose Grad-CAM for the CNN, the most faithful (deleting its most salient samples hurt the model most), and occlusion for the BiLSTM. Of four note generators tried on 50 beats, all made zero unsupported claims; the rule chose Gemini with worked examples and a fixed output schema, which covered every required fact in 2.6 s per note. Figure 8 shows the demonstrator for one S beat: the offline analysis by our models, which a review written by Gemini from those results then restates, claim by claim checked.

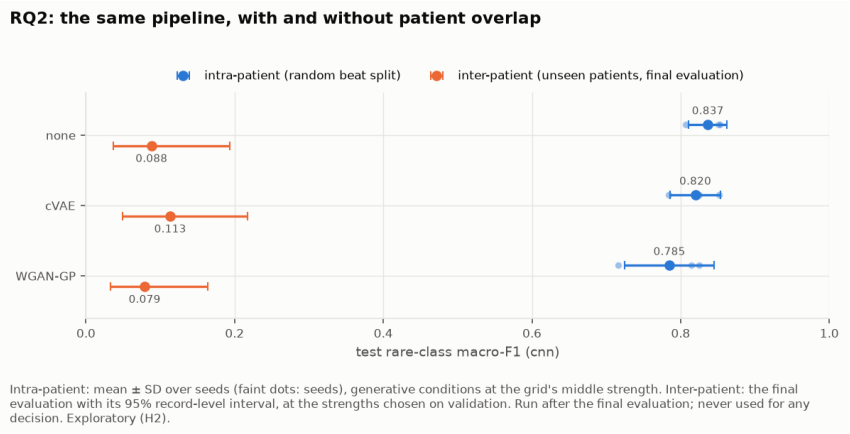


Figure 7: The same pipeline, scored on held-out beats of the training patients (blue) and on unseen patients (orange): the rare-class macro-F1 falls from about 0.8 to about 0.1, with or without generative augmentation.

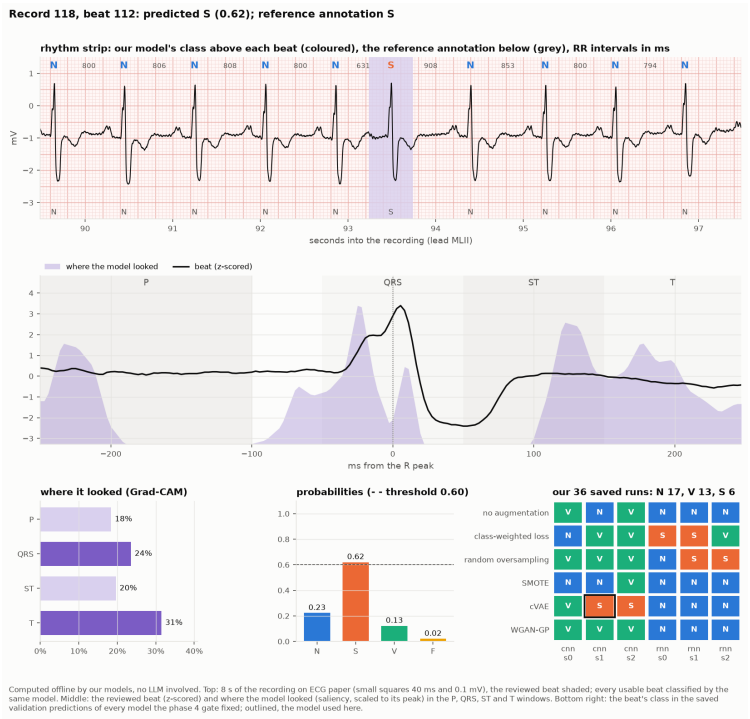


Figure 8: One S beat of a validation patient: the rhythm strip with each beat's predicted class and the reference label, where the model looked, the probabilities, and how all 36 saved runs classified the beat (only 6 called it S).

6 Discussion

Context before synthesis. With 17 training patients, the bottleneck for unseen patients is not the number of rare beats but the number of patients they come from. A generator trained on that set multiplies its dominant patient; what transferred instead was to read each beat against its own patient, as a cardiologist reads a rhythm strip: is the beat early for this heart, does it differ from this patient's normal beat, is the rhythm fibrillating? Once beats were expressed that way, the cVAE gave the best point estimate, which suggests the order in which the two ideas should be combined.

The protocol decides the score. The same network scores 0.837 or 0.088 depending only on whether the test patients were seen in training, so results on random beat splits cannot be ranked against inter-patient ones.

Limitations. The rare test beats sit in two patients, so intervals are wide and a single record can move the primary measure; the follow-up pipeline is post hoc and partly fitted to repeated test reads, which the unseen-patient check only partly corrects; and it uses more training patients than the published methods it is compared with. Fusion beats could not be checked on unseen patients. The notes restate the classifier and do

not diagnose.

7 Conclusion

Under a strict inter-patient protocol with rules fixed in advance, class-conditional generative augmentation did not detectably improve rare-arrhythmia detection, because the synthetic beats reproduce the few training patients they are learned from. Patient context did: patient-relative beats, prematurity against the patient's own rhythm, an atrial-fibrillation veto and ensembling raised the rare-class macro-F1 from 0.113 to 0.587, above classic inter-patient methods and level with the best recent one, ahead of it on fusion beats. Whether synthesis adds to context needs what it cannot manufacture: more patients with rare beats.

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